Matthew P. Maisonave

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PROFESSIONAL SUMMARY

Graduate computer science student and published researcher focused on GPU-accelerated software, high-performance computing, and computer architecture. Hands-on experience developing, profiling, and benchmarking CUDA and Triton kernels using shared-memory tiling, register blocking, coalesced memory access, kernel fusion, and sparse data structures. Experienced in performance analysis, numerical validation, automated testing, reproducible experimentation, and PTX/SASS inspection, with graduate study in high-performance computing and Fall 2026 coursework in computer architecture, reverse engineering, and data science.

EDUCATION

Old Dominion University (ODU) Norfolk, VA
M.S., Computer Science. *Expected December 2026*
Relevant Coursework: High-Performance Computing on Emerging Architectures, Bioinformatics, Artificial Intelligence, Computer Networks
Fall 2026 Coursework: Computer Architecture, Malware Analysis and Reverse Engineering, Introduction to Data Science and Analytics

Old Dominion University (ODU) Norfolk, VA
B.S., Computer Science. *December 2025*

SKILLS

GPU and Parallel Computing: CUDA, Triton, PyTorch, parallel algorithms, shared-memory tiling, coalesced memory access, kernel fusion, register reuse, workload partitioning, sparse computing, CSR, PTX/SASS inspection, CUDA event benchmarking, numerical validation
Programming Languages: C++, CUDA C++, Python, Rust, SQL, Java, Bash, JavaScript
Systems & Software Engineering: Linux/UNIX, Git, GitHub, Make, SSH, object-oriented programming, data structures and algorithms, memory management, API integration, automated testing, debugging, performance analysis
AI, Data, and Scientific Computing: NumPy, pandas, SciPy, machine learning, deep learning fundamentals, large language models, information retrieval, NLP, model evaluation, precision, recall, F1-score
Bioinformatics: Biopython, BLASTP, BLASTN, MMseqs2, MUMmer, CLUSTALW, sequence alignment, NCBI, Ensembl Plants, UniProt

EXPERIENCE

U-RISE Trainee, Bioinformatics Research May 2024 - May 2025
Old Dominion University Research Foundation *Norfolk, VA*
- Developed a reproducible Python bioinformatics pipeline using Biopython, pandas, and NumPy to integrate genomic, transcriptomic, and proteomic multi-omics datasets.
- Automated data ingestion, validation, feature extraction, annotation, and dataset generation for downstream machine learning experiments on understudied plant species.
- Executed BLASTP and BLASTN sequence alignment and homology searches; acquired and preprocessed data from NCBI, Ensembl Plants, and UniProt.

REU Intern May 2024 - August 2024
Old Dominion University Research Foundation *Norfolk, VA*
- Engineered Python experimental pipelines for scientific news verification using large language models and the GPT-3.5 API.
- Evaluated chain-of-thought, zero-shot, and few-shot prompting against ground-truth labels using precision, recall, and F1-score metrics.
- Automated experiment execution and results analysis with NumPy and Matplotlib; identified that zero-shot prompting outperformed chain-of-thought for stance labeling in the tested dataset.

Undergraduate Research Assistant August 2023 - September 2024
Old Dominion University Department of Computer Science *Norfolk, VA*
- Curated, cleaned, and validated ground-truth datasets used to benchmark multimodal search engines for scientific patent figures.
- Collaborated on annotation procedures and quality-assurance protocols to improve label consistency, data accuracy, and reproducibility.
- Contributed to research spanning information retrieval, natural language processing, machine learning, and large-scale data analysis.

PROJECTS

GPU-Accelerated Fused GEMM

C++, CUDA, Triton, Python, NumPy, Make

- Implemented CPU, naive CUDA, tiled CUDA, register-blocked CUDA, fused CUDA, and Triton versions of C = ReLU(αAB + bias), with automated numerical validation across square and rectangular workloads.
- Improved custom CUDA performance by up to 7.00× over naive fused CUDA and 5.21× over basic shared-memory tiling using 64×64×16 blocking, per-thread 4×4 register tiles, coalesced global loads, shared-memory staging, and register reuse.
- Benchmarked fused versus separate epilogue execution across seven workloads, reducing latency on 6/7 cases by up to 8.87%; inspected PTX/SASS and ptxas output, confirming 80 registers/thread, 8 KiB shared memory/block, zero spills, and generated LDG/STS/LDS/FFMA/FMNMX instruction paths.

GPU Cache Simulator

C++, Computer Architecture, Cache Modeling

- Built a configurable set-associative cache simulator with parameterized capacity, line size, and associativity, modeling 64-bit address decomposition, hits/misses, and write-back/write-allocate behavior.
- Implemented LRU replacement, dirty-line tracking, evictions, and writeback accounting using logical access timestamps and contiguous cache metadata storage with std::vector and zero-allocation std::span set views.
- Developed deterministic tests across direct-mapped and set-associative configurations to validate cache-line boundaries, spatial locality, replacement ordering, dirty-state persistence, writebacks, and per-set isolation.

Sparse Matrix-Vector Multiplication on GPU

Python, Triton, PyTorch, SciPy, CSR

- Developed and validated a custom Triton CSR sparse matrix-vector multiplication kernel against manual CPU CSR, SciPy, and PyTorch sparse-CSR implementations across workloads up to 1M rows and 8M nonzeros.
- Built GPU-resident benchmarking infrastructure using CUDA-event timing, warmups, repeated measurements, numerical validation, and configurable matrix size and row density; custom Triton SpMV reached 7.29× speedup over SciPy CPU at 1M rows / 8M nonzeros.
- Characterized sparse workload scaling from 4–64 nonzeros/row, with a repeat-verified 17.75× Triton speedup over SciPy at 64 nonzeros/row and strong workload-density dependence in the fixed row-parallel kernel.

CUDA Stencil Memory Optimization

C++, CUDA, Linux, HPC

- Implemented and validated naive and shared-memory 1D and 2D CUDA stencils with CPU references, runtime-radius halo handling, dynamic shared memory, partial-block safety, configurable launch geometry, and CUDA-event benchmarking.
- Built a reproducible 35-configuration benchmark pipeline with CSV output and generated performance figures; swept 1D radii through 16 across 32/64/128/256-thread blocks and 2D workloads from 256^2–4096^2 across 8×8, 16×16, and 32×32 blocks.
- Found shared-memory staging faster in only 4/20 1D configurations and slower in 15/15 2D configurations (2048^2, 16×16: 0.0386 ms shared vs. 0.0246 ms naive), demonstrating how halo loading and synchronization overhead can offset explicit reuse when regular/coalesced accesses already exhibit strong locality.

SELECTED PUBLICATIONS

- “Comparative analysis of six new chloroplast genomes in Platanthera (Orchidaceae) enhances understanding of its diversification.” Scientific Reports, 16, 24675 (2026).
- "SciTableQA: A Question-Answering Benchmark for Complex Scientific Tables." 29th International Conference on Theory and Practice of Digital Libraries (TPDL 2025), Tampere, Finland.
- "Humans vs. LLMs on Open Domain Scientific Claim Verification: A Baseline Study." 2nd International Workshop on Artificial Intelligence for the Science of Science (AI4SciSci 2025).

AWARDS & HONORS

U-RISE Scholar | Dean's List (3x) | Dean's Perfect Scholar (3x) | $9,500 in university scholarships